



Decibels

112-BOLCEW05

Audiometry

UNCLASSIFIED



UNCLASSIFIED

Terminal Learning Objective



UNCLASSIFIED

The student will solve problems using decibels principles in a classroom environment, given a networked computer workstation, training aids, and appropriate references by demonstrating understanding of power ratios, power levels dBw and dBm, rules for manipulation, relationship between dBm and dBw, gain and loss to support the understanding of Electronic Warfare Math critical to EW effects, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

UNCLASSIFIED

Learning Step Activities



UNCLASSIFIED

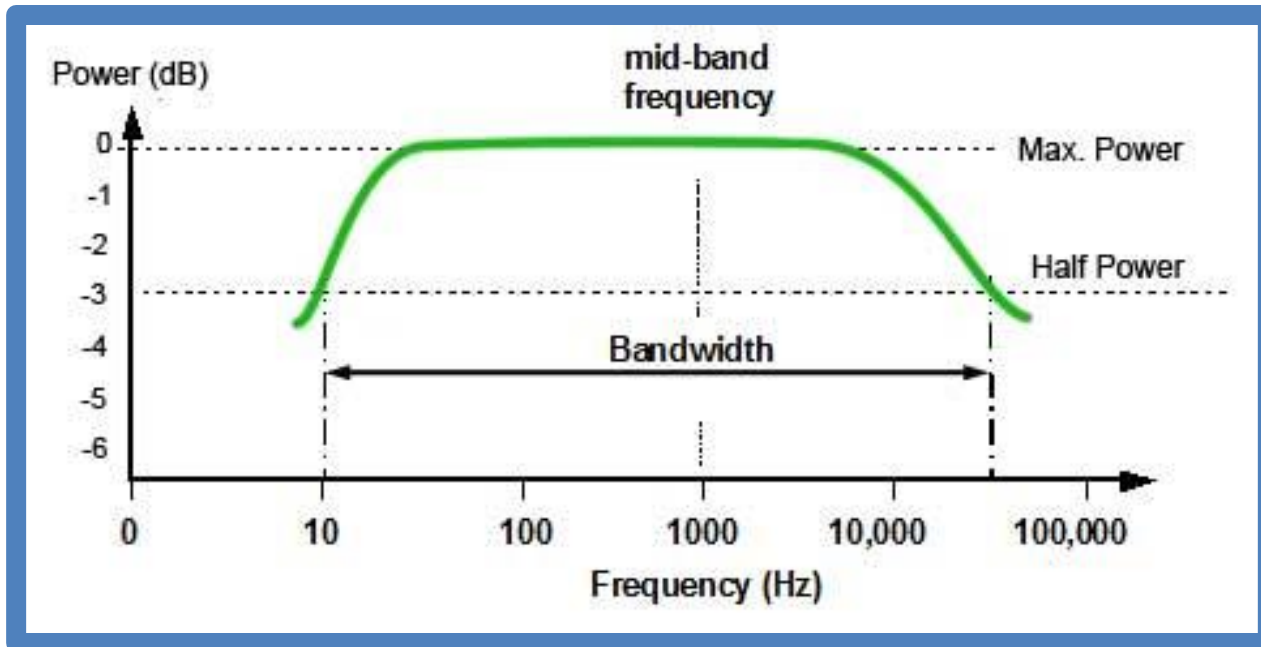
- Define Power ratios
- Define Decibels
- Define Power levels dBw and dBm
- Define Rules for manipulation
- Define Relationship between dBm and dBw
- Define Gain and Loss
- Define Systems with multiple gains and losses
- Decibels Practical Exercise

UNCLASSIFIED

Power Ratios



UNCLASSIFIED



UNCLASSIFIED

Power Ratios (1 of 2)



UNCLASSIFIED

Power ratio - is a number, $\frac{P_{out}}{P_{in}}$, which describes the relative difference between two power levels, P_{out} and P_{in}

Example:

If $P_{out} = 4$ watts and $P_{in} = 2$ watts, what is the power ratio?

- $\frac{P_{out}}{P_{in}} = \frac{4 \text{ watts}}{2 \text{ watts}} = 2$. So P_{out} is twice as strong as P_{in}

UNCLASSIFIED

Power Ratios (2 of 2)



UNCLASSIFIED

Example:

If $P_{\text{out}} = 6$ watts and the power ratio is 0.5, what is P_{in} ?

$$\frac{6 \text{ Watts}}{P_{\text{in}}} = 0.5$$

$$6 \text{ watts} = 0.5(P_{\text{in}})$$

$$\frac{6 \text{ watts}}{0.5} = P_{\text{in}}$$

$$\mathbf{12 \text{ watts}} = P_{\text{in}}$$

UNCLASSIFIED



Questions?

Decibels



UNCLASSIFIED

Decibel Scale

eps10



UNCLASSIFIED

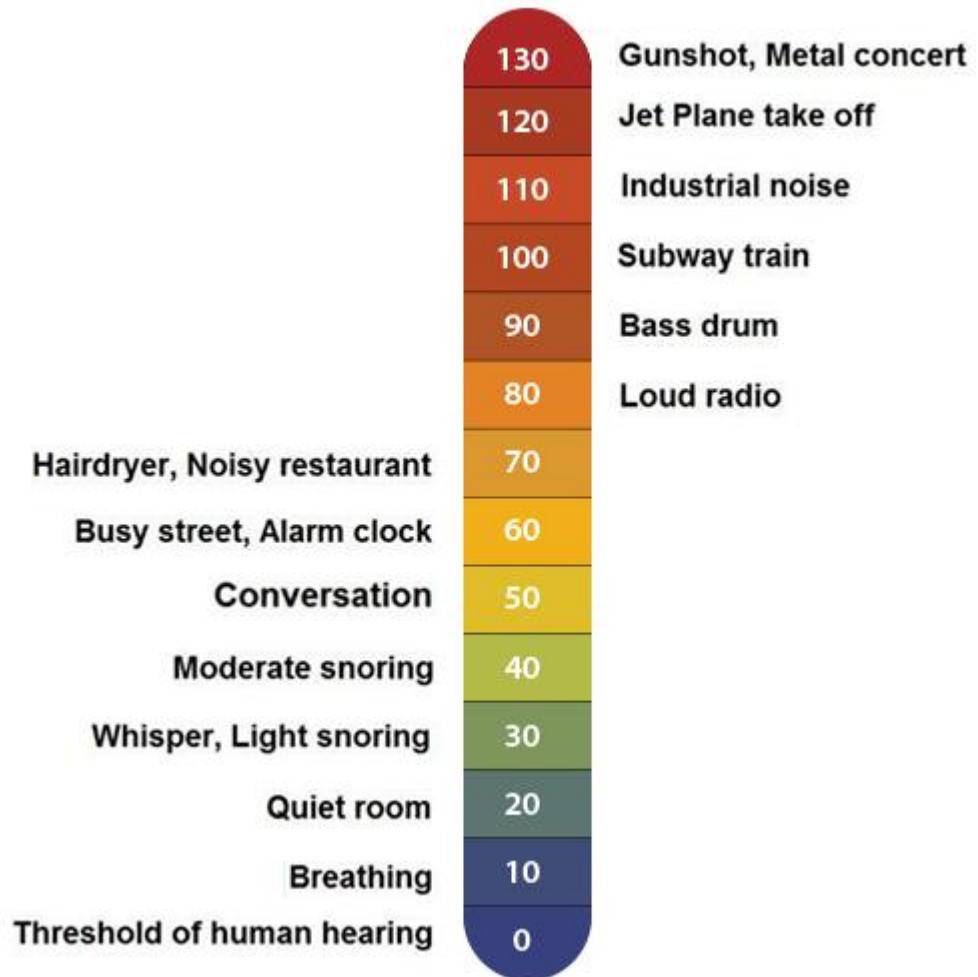
Decibels (1 of 11)



UNCLASSIFIED

Power ratios are described on a logarithmic scale.

The scale is called the decibel (dB)



UNCLASSIFIED

Decibels (2 of 11)



UNCLASSIFIED

A power ratio may be expressed in numerical form, or in logarithmic form

A power ratio expressed in decibel form will always be followed by the dB symbol

When no dB symbol is present, a power ratio will be understood to be in numerical form

UNCLASSIFIED

Decibels (3 of 11)



UNCLASSIFIED

Decibels (dB) defined: Given the power ratio $\frac{P_{out}}{P_{in}}$ write

$\frac{P_{out}}{P_{in}} \leftrightarrow (10 \log_{10} \frac{P_{out}}{P_{in}}) \text{ dB}$ and read “the power ratio $\frac{P_{out}}{P_{in}}$ in numerical form corresponds to $(10 \log_{10} \frac{P_{out}}{P_{in}})$ decibels in dB or logarithmic form

Example: If $\frac{P_{out}}{P_{in}} = 2$, then $2 \leftrightarrow \underline{\hspace{2cm}}$ dB.

$$\frac{P_{out}}{P_{in}} = 2 \leftrightarrow \text{to } (10 \log_{10} 2) \text{ dB} = (10 \times 0.3) \text{ dB} = 3 \text{ dB}$$

So, $2 \leftrightarrow 3 \text{ dB}$, or a power ratio of 2 corresponds to **3 dB**

UNCLASSIFIED

Decibels (4 of 11)



UNCLASSIFIED

Example: If $\frac{P_{out}}{P_{in}} = 4$, then $4 \leftrightarrow$ _____ dB

$$\begin{aligned}\frac{P_{out}}{P_{in}} = 4 &\leftrightarrow \text{to } (10 \log_{10} 4) \text{ dB} \\ &= (10 \times 0.6) \text{ dB} \\ &= \mathbf{6 \text{ dB}}\end{aligned}$$

Example: $8 \leftrightarrow$ _____ dB

$$\begin{aligned}8 &\leftrightarrow \text{to } (10 \log_{10} 8) \text{ dB} \\ &= (10 \times 0.9) \text{ dB} \\ &= \mathbf{9 \text{ dB}}\end{aligned}$$

UNCLASSIFIED

Decibels (5 of 11)



UNCLASSIFIED

$2 \leftrightarrow 3 \text{ dB}$

$4 \leftrightarrow 6 \text{ dB}$

$8 \leftrightarrow 9 \text{ dB}$

What is noticed about the power ratio compared to its decibel equivalent?

If a power ratio increases by a factor of 2, then the corresponding dB form is increased by the sum of about 3 dB

UNCLASSIFIED

Decibels (6 of 11)



UNCLASSIFIED

Example:

Given that a power ratio of 5 $\leftrightarrow$ 7 dB, 10 $\leftrightarrow$ _____ dB and 20 $\leftrightarrow$ _____ dB

10 $\leftrightarrow$ **10**

20 $\leftrightarrow$ **13**

UNCLASSIFIED

Decibels (7 of 11)



UNCLASSIFIED

Multiplying by a power ratio in numerical form by 2 is equivalent to adding 3 dB to its dB form

Multiplying the numerical power ratio by $\frac{1}{2}$ (or dividing by 2) is equivalent to subtracting 3 dB from its dB form

UNCLASSIFIED

Decibels (8 of 11)



UNCLASSIFIED

Example:

Given that $28 \leftrightarrow 14.5 \text{ dB}$, $14 \leftrightarrow \underline{\hspace{2cm}} \text{ dB}$

$14 \leftrightarrow 11.5 \text{ dB}$

UNCLASSIFIED

Decibels (9 of 11)



UNCLASSIFIED

Note:

$$\begin{aligned} 10 &\leftrightarrow (10 \log_{10} 10) \text{dB} \\ &= (10 \times 1) \text{dB} \\ &= 10 \text{ dB, so } 10 \leftrightarrow 10 \text{ dB} \end{aligned}$$

Multiplying (dividing) a numerical power ratio by 10 is equivalent to adding (subtracting) 10 dB to (from) its dB value

$$\begin{aligned} \frac{1}{10} &\leftrightarrow [10 \log \frac{1}{10}] \text{dB} \\ &= ((10 \log_{10} 1) - (10 \log_{10} 10)) \text{dB} \\ &= (0 - 10) \text{dB} \\ &= -10 \text{ dB, so } \frac{1}{10} \leftrightarrow -10 \text{ dB} \end{aligned}$$

UNCLASSIFIED

Decibels (10 of 11)



UNCLASSIFIED

Since $\frac{P_1}{P_2} \leftrightarrow (10 \log_{10} \frac{P_{out}}{P_{in}}) \text{ dB}$, it follows that

$$N \text{ dB} \leftrightarrow 10^{\frac{N}{10}} = \log^{-1} \frac{N}{10} = \text{anti log } \frac{N}{10}$$

(Antilog may be the 10^x button on the calculator.)

UNCLASSIFIED

Decibels (11 of 11)



UNCLASSIFIED

Example: Convert 46 dB to a power ratio format

$$\begin{aligned} N \text{ dB} &\leftrightarrow 10^{\frac{N}{10}}, \text{ so } 46 \text{ dB} \leftrightarrow 10^{\frac{46}{10}} \\ &= 10^{4.6} \\ &\cong 39,810.72 \cong 40,000 \\ \mathbf{46 \text{ dB} \cong 40,000} \end{aligned}$$

Example: Convert -13 dB to a power ratio format

$$\begin{aligned} N \text{ dB} &\leftrightarrow 10^{\frac{N}{10}}, \text{ so } -13 \text{ dB} \leftrightarrow 10^{\frac{-13}{10}} \\ &= 10^{-1.3} \\ &\cong 0.05 \\ \mathbf{-13 \text{ dB} \cong 0.05} \end{aligned}$$

UNCLASSIFIED



Questions?



UNCLASSIFIED

Decibel (dB)

What is dB, dBm, dBW and dBV ?

Power Levels dBw and dBm (1 of 3)



UNCLASSIFIED

Decibel notation is used to express the relationship between a random power level and a fixed reference power level

- Two reference levels commonly used: 1 watt and 1 mill watt
- When the watt(s) scale is used, the unit of measure is the dBw
- When the milliwatt (mw) scale is used, the unit of measure is the dBm

UNCLASSIFIED

Power Levels dBw and dBm (2 of 3)



UNCLASSIFIED

$$\begin{aligned} P \text{ watts} &\leftrightarrow (10 \log \frac{P \text{ watts}}{1 \text{ watt}}) \text{dBw} \\ &= [(10 \log P) - (10 \log 1)] \text{dBw} \\ &= [(10 \log P) - (0)] \text{dBw} \\ &= (10 \log P) \text{dBw} \end{aligned}$$

$$\begin{aligned} P \text{ milliwatts} &\leftrightarrow (10 \log \frac{P \text{ milliwatts}}{1 \text{ milliwatt}}) \text{dBm} \\ &= [(10 \log P) - (10 \log 1)] \text{dBm} \\ &= [(10 \log P) - (0)] \text{dBm} \\ &= (10 \log P) \text{dBm} \end{aligned}$$

Equals =

$$P \text{ watts} = (10 \log P) \text{dBw}$$

$$P \text{ milliwatts} = (10 \log P) \text{dBm}$$

UNCLASSIFIED

Power Levels dBw and dBm (3 of 3)



UNCLASSIFIED

Example:

A signal has a power level of 500 milliwatts. What is the level in dBm?

$$\begin{aligned} 500 \text{ milliwatts} &\leftrightarrow (10 \log 500) \text{dBm} \\ &= (10 \log 500) \text{dBm} \\ &= (10 \times 2.6990) \text{dBm} \\ &= \mathbf{26.99 \text{ dBm}} \end{aligned}$$

UNCLASSIFIED



Questions?

Rules for Manipulation



UNCLASSIFIED

***the language that the field
of mathematics uses to talk
about the abstract world of numbers***

UNCLASSIFIED

Rules for Manipulation (1 of 2)



UNCLASSIFIED

Rules for manipulating dB:

- Adding non-suffixed dB (dB without a reference, unlike dB_w and dB_m) to (or subtracting them from) dBm (or dBw) does not change the power reference. The result is still in dBm (or dBw).
- dBm may be subtracted from dBm. The result is in dB. dBw may be subtracted from dBw. The result is also in dB.
- To subtract dBm from dBw or vice versa, you must change the reference level on one of them (will be covered later)
- Do not add dBm or dBw to themselves or to each other.

UNCLASSIFIED

Rules for Manipulation (2 of 2)



UNCLASSIFIED

Examples:

$$10 \text{ dBm} - 3 \text{ dB} \\ = 7 \text{ dBm}$$

$$24 \text{ dBw} + 6 \text{ dB} \\ = 30 \text{ dBw}$$

$$15 \text{ dBm} - 5 \text{ dBm} \\ = 10 \text{ dB}$$

$$17 \text{ dBw} - 8 \text{ dBw} \\ = 9 \text{ dB}$$

$$15 \text{ dBw} - 3 \text{ dBm} \\ = 45 \text{ dBm} - 3 \text{ dBm} = 42 \text{ dBm} \text{ (will be covered later in the lesson)}$$

UNCLASSIFIED

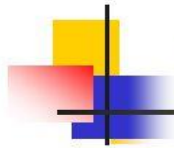


Questions?

Relationship Between dBm and dBw



UNCLASSIFIED



dBW and dBm

- $\text{POWER}_{\text{dBW}} = 10 \log (\text{POWER}_W / 1W)$
 - $0 \text{ dBW} = 1W$
- $\text{POWER}_{\text{dBm}} = 10 \log (\text{POWER}_{\text{mW}} / 1\text{mW})$
 - $0 \text{ dBm} = 1\text{mW}$
- $+30 \text{ dBm} = 0 \text{ dBW}$
- $0 \text{ dBm} = -30 \text{ dBW}$

UNCLASSIFIED

Relationship Between dBm and dBw (1 of 2)



UNCLASSIFIED

By definition, 1 milliwatt = $\frac{1 \text{ watt}}{1000}$, therefore N' milliwatts = $\frac{N' \text{ watts}}{1000}$

Taking 10 times the log of both sides, we have:

- $N \text{ dBw} = N \text{ dBm} - 30 \text{ dB}$
- $N \text{ dBw} = (N - 30) \text{ dBm}$
- This is the general rule for converting dBm to dBw.

UNCLASSIFIED

Relationship Between dBm and dBw (2 of 2)



UNCLASSIFIED

Watts	↔	dBw	=	dBm	↔	Milli watts	=	Watts
10^6	↔	60	=	90	↔	10^9	=	10^6
10^5	↔	50	=	80	↔	10^8	=	10^6
10^4	↔	40	=	70	↔	10^7	=	10^6
10^3	↔	30	=	60	↔	10^6	=	10^6

UNCLASSIFIED

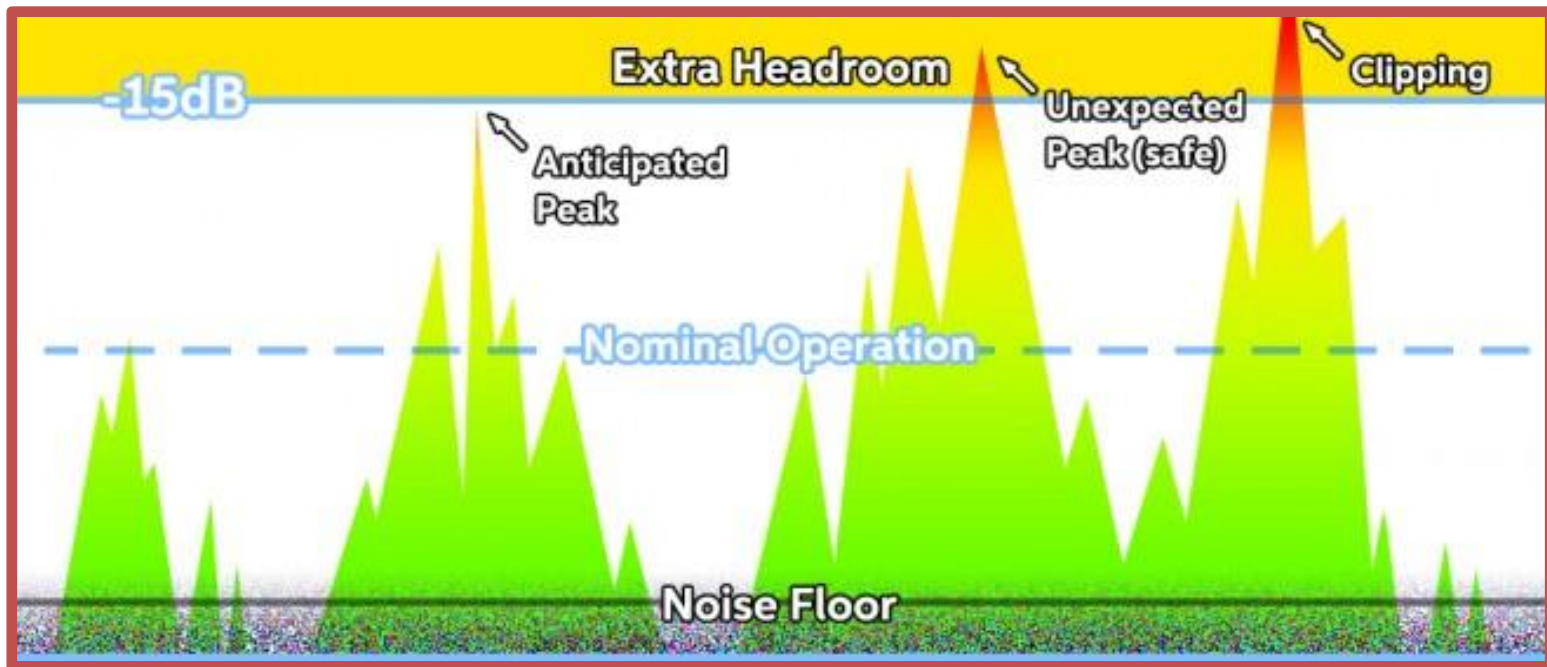


Questions?

Gain and Loss



UNCLASSIFIED



UNCLASSIFIED

Gain and Loss (1 of 6)



UNCLASSIFIED

- A common use of dB is to describe a gain of an amplifier.
- In general, gain is defined as $g = \frac{P_{out}}{P_{in}}$.
- The gain in dB is $g \text{ dB} = [10 \log(\frac{P_{out}}{P_{in}})]\text{dB}$.

UNCLASSIFIED

Gain and Loss (2 of 6)



UNCLASSIFIED

Example: If one watt of power was fed into an amplifier and 100 watts appeared as its output, then the amplifier would have multiplied the input by 100.

$$P_{in} = 1w \rightarrow g \rightarrow P_{out} = 100w$$

The device's gain, g , is then defined as:

$$g = \frac{P_{out}}{P_{in}} = \frac{100 w}{1 w} = 100$$

The gain in dB is:

$$g \text{ dB} = (10 \log 100) \text{ dB} = 20 \text{ dB}$$

UNCLASSIFIED

Gain and Loss (3 of 6)



UNCLASSIFIED

Example: An amplifier has a gain of 350. What is its gain in dB?

$$P_{in} \rightarrow g = 350 \rightarrow P_{out}$$

$$g \leftrightarrow (10 \log g) \text{ dB}$$

$$= [(10 \log 350)] \text{ dB}$$

$$= (10 \times 2.5441) \text{ dB}$$

$$= \mathbf{25.44 \text{ dB}}$$

UNCLASSIFIED

Gain and Loss (4 of 6)



UNCLASSIFIED

If for some non-amplifying device, the P_{in} is greater than the P_{out} , a power loss takes place.

- The symbol for loss is the lowercase Greek letter, alpha (α)

Define the loss, α , of such a device to be $\alpha = \frac{P_{in}}{P_{out}}$

- therefore $\alpha = \frac{1}{g}$.
- In decibel form $\alpha \text{ dB} = -g \text{ dB}$. In other words, loss is the inverse of gain.

UNCLASSIFIED

Gain and Loss (5 of 6)



UNCLASSIFIED

Example: Find the loss in dB of the following device.

$$P_{in} = 10 \text{ watts} \rightarrow \alpha \rightarrow P_{out} = 1 \text{ watt}$$

$$\begin{aligned} \alpha \text{ dB} &= -g \text{ dB} \\ &= -[10 \log(\frac{P_{out}}{P_{in}})] \text{ dB} \end{aligned}$$

$$= -[10 \log [\frac{1w}{10w}]] \text{ dB}$$

$$= -[10 \log(0.1)] \text{ dB}$$

$$= - [10 \times (-1)] \text{ dB}$$

$$= \mathbf{10 \text{ dB}}$$

UNCLASSIFIED

Gain and Loss (6 of 6)



UNCLASSIFIED

Example: Find the loss in dB of the following device.

$$P_{in} = 10 \text{ watts} \rightarrow \alpha \rightarrow P_{out} = 1 \text{ watt}$$

The calculation could also be done as follows:

$$\alpha \text{ dB} = [10 \log(\frac{P_{in}}{P_{out}})] \text{ dB}$$

$$= [10 \log [\frac{10 \text{ w}}{1 \text{ w}}]] \text{ dB}$$

$$= [10 \log(10)] \text{ dB}$$

$$= [10 \times (1)] \text{ dB}$$

$$= 10 \text{ dB}$$

UNCLASSIFIED

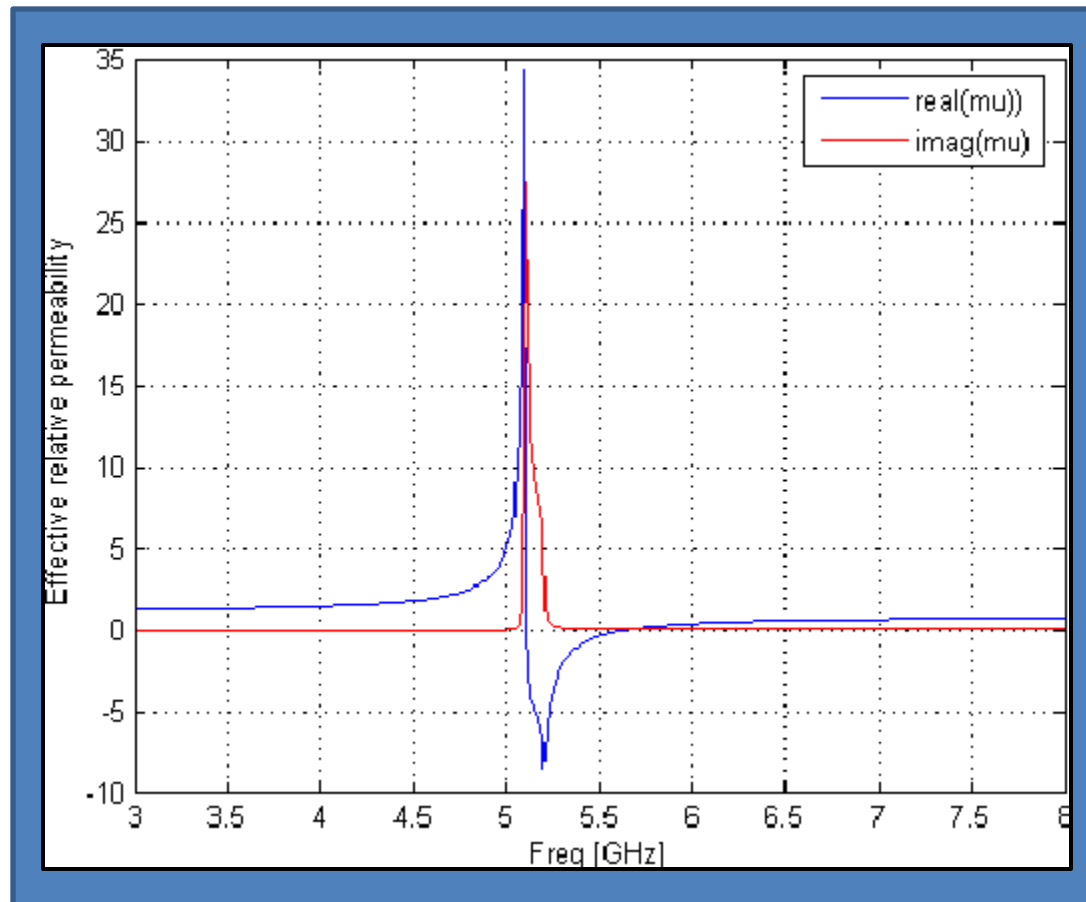


Questions?

Systems with Multiple Gains and Losses



UNCLASSIFIED



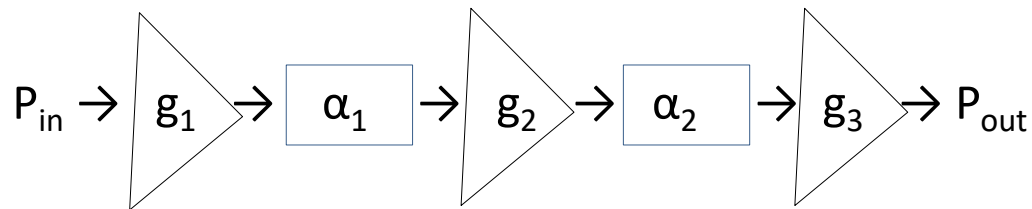
UNCLASSIFIED

Systems with Multiple Gains and Losses (1 of 5)



UNCLASSIFIED

In systems with multiple sources of gains and losses one behind another, the overall system gain can be determined and can replace the group of individual gains and losses



$$P_{out} = P_{in} \times \frac{g_1 \times g_2 \times g_3}{\alpha_1 \times \alpha_2}$$

$$g_{SYSTEM} = \frac{P_{out}}{P_{in}} = \frac{g_1 \times g_2 \times g_3}{\alpha_1 \times \alpha_2}$$

$$\text{So } P_{in} \rightarrow g_{SYSTEM} \rightarrow P_{out}$$

UNCLASSIFIED

Systems with Multiple Gains and Losses (2 of 5)



UNCLASSIFIED

Example:

If $g_1 = 500$, $g_2 = 100$, $g_3 = 10$, $\alpha_1 = 20$, and $\alpha_2 = 5$, what is the system gain?

$$g_{\text{SYSTEM}} = \frac{g_1 \times g_2 \times g_3}{\alpha_1 \times \alpha_2}$$

$$= \frac{500 \times 100 \times 10}{20 \times 5}$$

$$= \mathbf{5000}$$

UNCLASSIFIED

Systems with Multiple Gains and Losses (3 of 5)



UNCLASSIFIED

Example:

If $g_1 = 500$, $g_2 = 100$, $g_3 = 10$, $\alpha_1 = 20$, and $\alpha_2 = 5$, what is the system gain?

In decibel notation:

$$g_{\text{SYSTEM}} \text{ dB} = g_1 \text{ dB} + g_2 \text{ dB} + g_3 \text{ dB} - \alpha_1 \text{ dB} - \alpha_2 \text{ dB}$$

$$= (10 \log g_1) \text{ dB} + (10 \log g_2) \text{ dB} + (10 \log g_3) \text{ dB} - (10 \log \alpha_1) \text{ dB} - (10 \log \alpha_2) \text{ dB}$$

$$= (10 \log 500) \text{ dB} + (10 \log 100) \text{ dB} + (10 \log 10) \text{ dB} - (10 \log 20) \text{ dB} - (10 \log 5) \text{ dB}$$

$$= 26.99 \text{ dB} + 20 \text{ dB} + 10 \text{ dB} - 13.01 \text{ dB} - 6.99 \text{ dB}$$

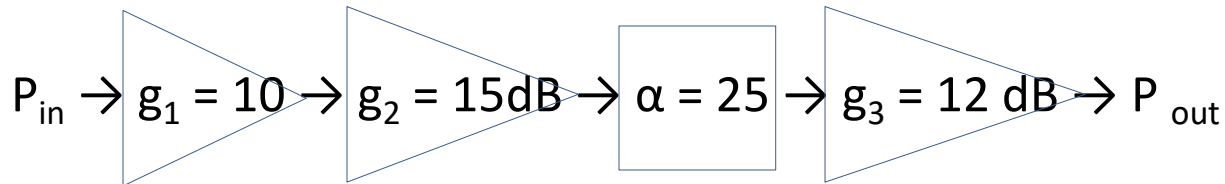
$$= \mathbf{36.99 \text{ dB}}$$

Systems with Multiple Gains and Losses (4 of 5)



UNCLASSIFIED

Example: In the following system, if $P_{in} = 10$ dBw, what is P_{out} ?



$$g_1 = 10 \leftrightarrow 10\text{ dB}$$

$$g_2 = 15\text{ dB}$$

$$g_3 = 12\text{ dB}$$

$$\alpha = [10 \log(25)]\text{dB}$$

$$= [10(1.3979)]\text{dB}$$

$$\cong 13.98\text{ dB}$$

$$g_{\text{SYSTEM}}\text{ dB} = g_1\text{ dB} + g_2\text{ dB} + g_3\text{ dB} - \alpha\text{ dB}$$

$$= (10 + 15 + 12 - 13.98)\text{ dB}$$

$$= 23\text{ dB}$$

UNCLASSIFIED

Systems with Multiple Gains and Losses (5 of 5)



UNCLASSIFIED

Then simplify it as:

$$P_{in} = 10 \text{ dBw} \rightarrow g_{SYSTEM} = 23 \text{ dB} \rightarrow P_{out}$$

$$P_{out} \text{ dBw} = P_{in} \text{ dBw} + g_{SYSTEM} \text{ dB}$$

$$= 10 \text{ dBw} + 23 \text{ dB}$$

$$= 33 \text{ dBw}$$

UNCLASSIFIED



Questions?

Summary



UNCLASSIFIED

During this lesson we have discussed:

- Define Power ratios
- Define Decibels
- Define Power levels dBw and dBm
- Define Rules for manipulation
- Define Relationship between dBm and dBw
- Define Gain and Loss
- Define Systems with multiple gains and losses

UNCLASSIFIED

References



UNCLASSIFIED

- Marthaler, Laura and Wytko, Holly Wytko, ed., MATH for SIGINT (2007), 12-1 through 12-14

UNCLASSIFIED